

USER GUIDE

Student Performance Tracker

User Guide

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1. Introduction

The **Student Performance Tracker** is a web-based application developed using **Python, Flask, and SQLite**. It helps users manage student records, store subject-wise grades, and generate reports such as average marks, subject toppers, and class averages. The application provides an easy-to-use interface for performing all student management tasks.

2. Technologies Used

- Python
- Flask
- SQLite
- HTML
- CSS

3. Features

The application provides the following features:

- Add Student
- Add Grades
- View Students
- Search Student
- Average Report
- Subject-Wise Topper
- Class Average
- Backup Student Data

4. How to Use the Application

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4.1 Home Page

After starting the application, the home page displays all available options.

Users can select any feature by clicking the corresponding button.

4.2 Add Student

Steps

1. Click **Add Student**.
2. Enter the Roll Number.
3. Enter the Student Name.
4. Click **Add Student**.

Example

Roll Number	Name
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101	Rahul
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The student record is successfully stored in the database.

4.3 Add Grades

Steps

1. Click **Add Grades**.
2. Enter the Roll Number.
3. Enter the Subject Name.
4. Enter the Marks.
5. Click **Save Grades**.

Example

Roll Number	Subject	Marks
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101	Mathematics	95
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The grades are stored successfully.

4.4 View Students

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Displays all students available in the database.

The students are displayed in **ascending order of Roll Number**.

4.5 Search Student

Steps

1. Click **Search Student**.
2. Enter the Roll Number.
3. Click **Search**.

If the student exists, the student's details and grades are displayed.

If the Roll Number does not exist, an appropriate message is shown.

4.6 Average Report

Displays the average marks of students based on the stored grades.

This helps users analyze student performance.

4.7 Subject-Wise Topper

Steps

1. Click **Subject Topper**.
2. Enter the Subject Name.
3. Click **Find Topper**.

The application displays:

- Roll Number
- Student Name
- Subject
- Highest Marks

4.8 Class Average

Steps

1. Click **Class Average**.

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2. Enter the Subject Name.
3. Click **Find Average**.

The application calculates and displays the average marks of the class for the selected subject.

4.9 Backup Data

Click **Backup Data**.

The application creates a text file containing all student records and grades for backup purposes.

5. Database

The application uses an SQLite database named:

students.db

The database contains the following tables:

Students Table

Column

Roll Number

Name

Grades Table

Column

Roll Number

Subject

6. Conclusion

The **Student Performance Tracker** provides an efficient and user-friendly way to manage student records and performance reports. It supports adding students, assigning grades, viewing reports, searching students, identifying subject toppers, calculating class averages, and creating backups of student data.

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The project demonstrates the use of **Flask**, **SQLite**, **HTML**, and **CSS** to build a complete web application.